

Multi-Objective Optimization - Report 1

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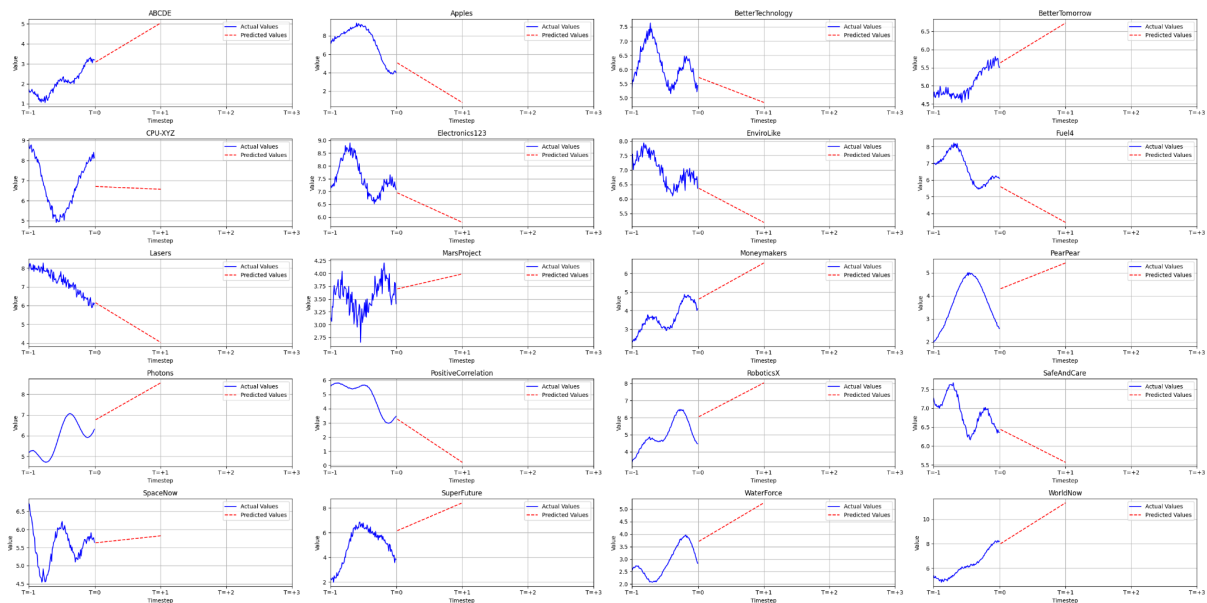
1. Problem Definition

The presented problem concerns multi-objective optimization for investment portfolio optimization used in the context of a simple “Portfolio Game”. In this particular implementation, the focus was on the WSM and ECM methods used in the context of this task. The “Portfolio Game” is a project based on predicting stock prices for 20 fictitious companies and utilizing Markovitz to create a Pareto front of non-dominated solutions that balance exploration and exploitation. To achieve this task three components must be implemented: first, a proper regression model must be implemented to predict companies' prices at future time points, second the WSM and ECM algorithms must be implemented to approximate the Pareto front.

2. Regression Models

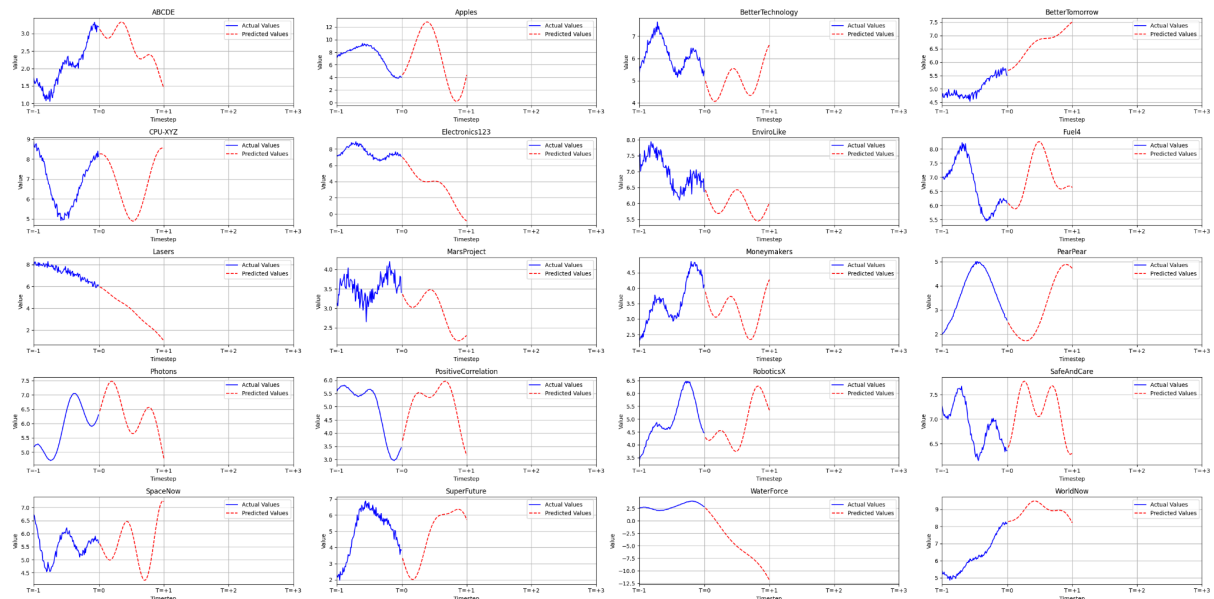
The task of regression posed a significant challenge in this project due to the lack of the data to properly train the models/fit the curves. Despite that, several methods were implemented with this short section focusing on visualizations and brief explanations of some of the approaches that turned out to be the most promising.

2.1 Linear Regression



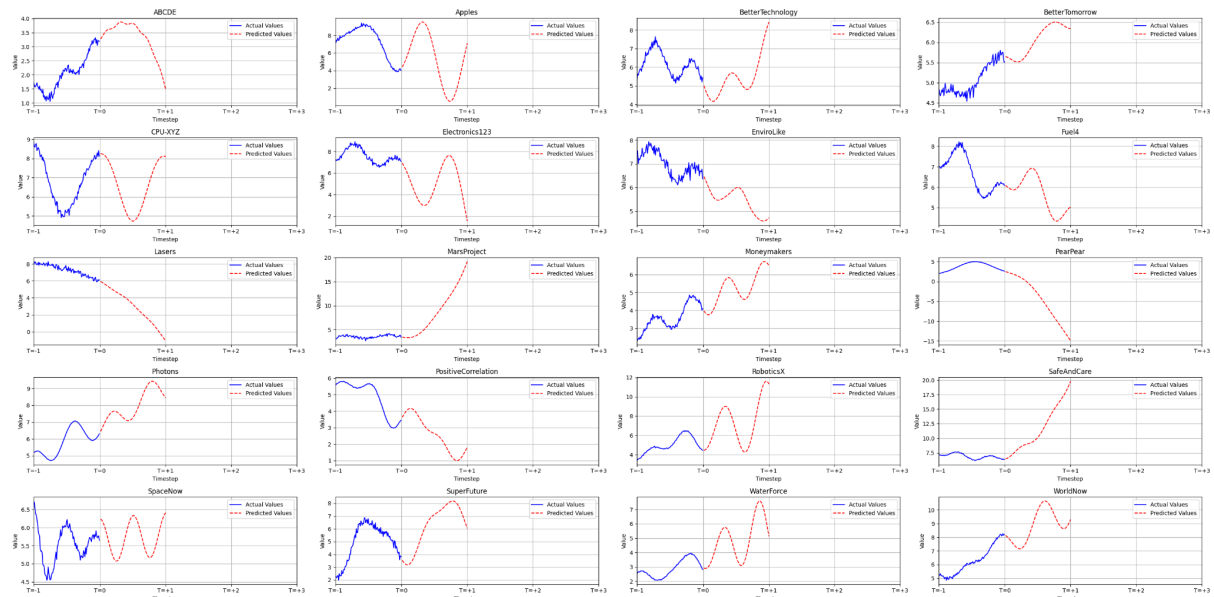
2.2 Sinusoidal Regression

This solution was based on fitting a curve defined as $A1 * \sin(B1 * x + C1) + A2 * \sin(B2 * x + C2) + D$ to the data.



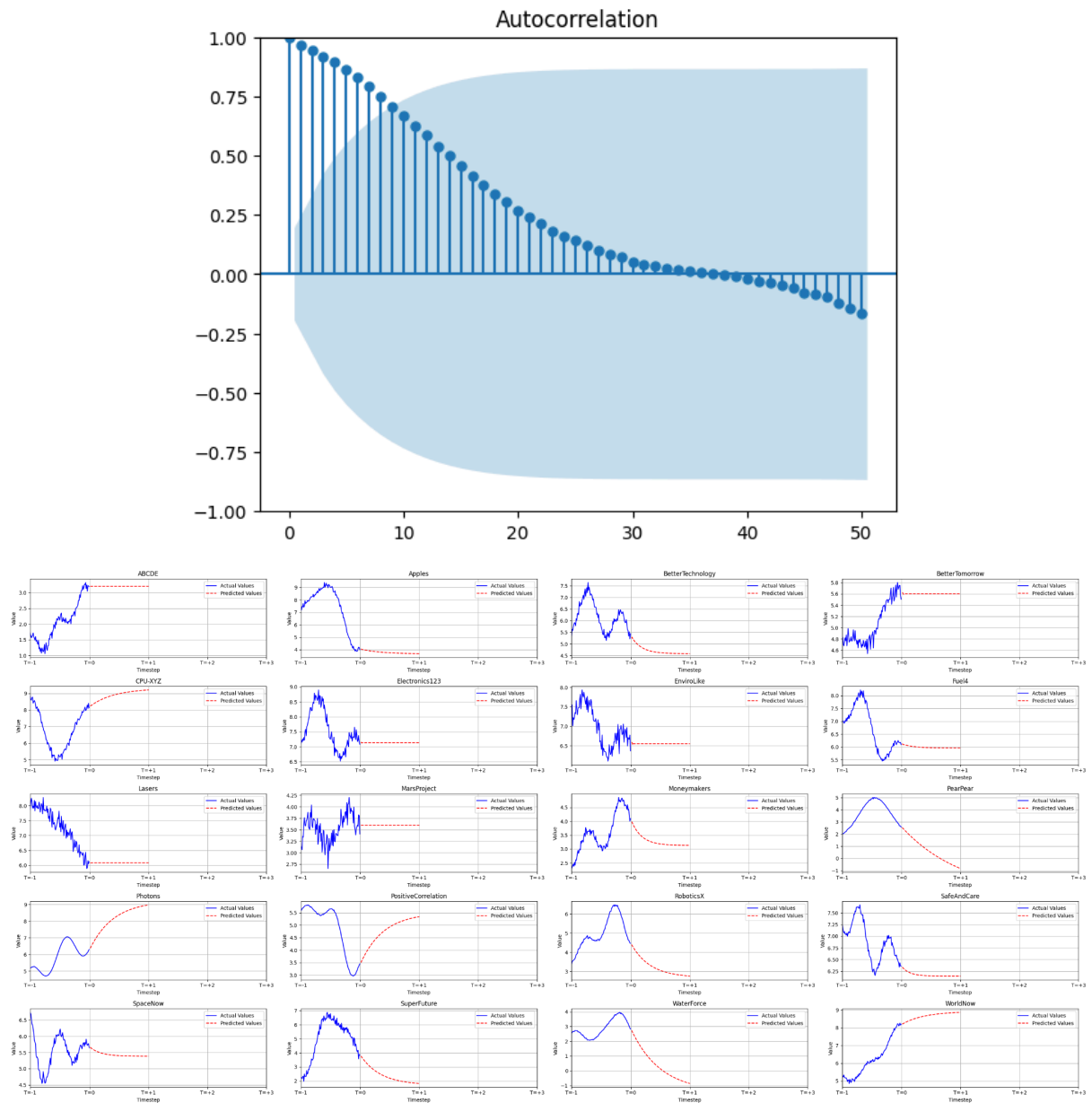
2.3 Complex Sinusoidal Regression

A more complex version of the previous solution, defined as fitting the curve $a * x + A1 * \sin(B1 * x + C1) + A2 * \sin(B2 * x + C2) + D$ to the data.

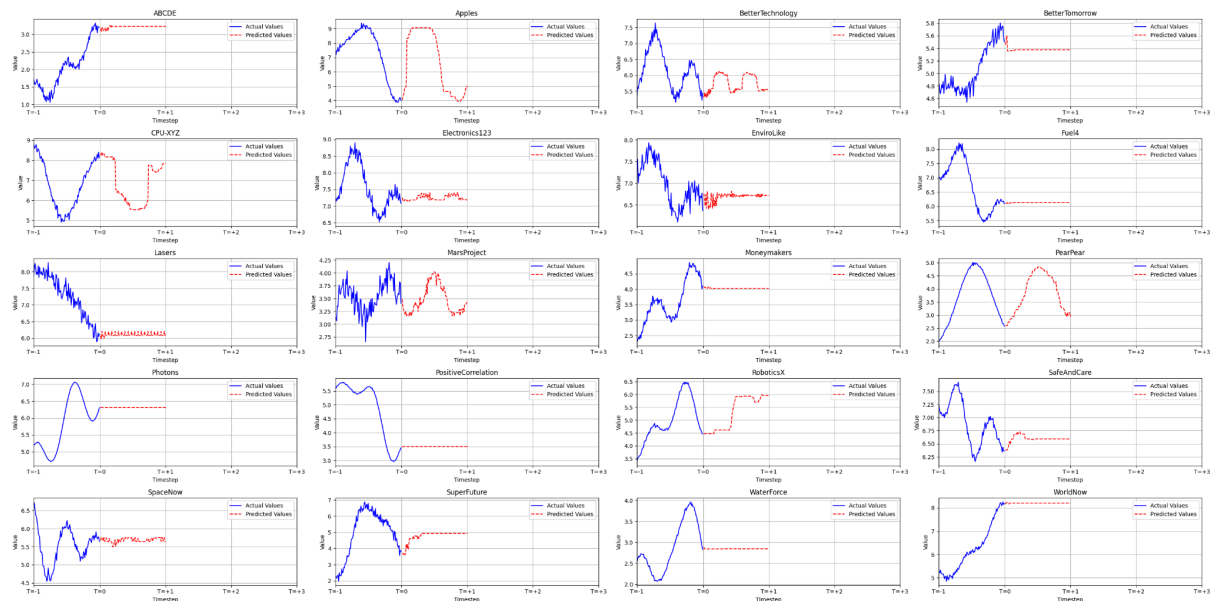


2.4 ARIMA

This solution is based on the well-known ARIMA method with the parameters being set by hand based on the autocorrelation and partial autocorrelation plots.

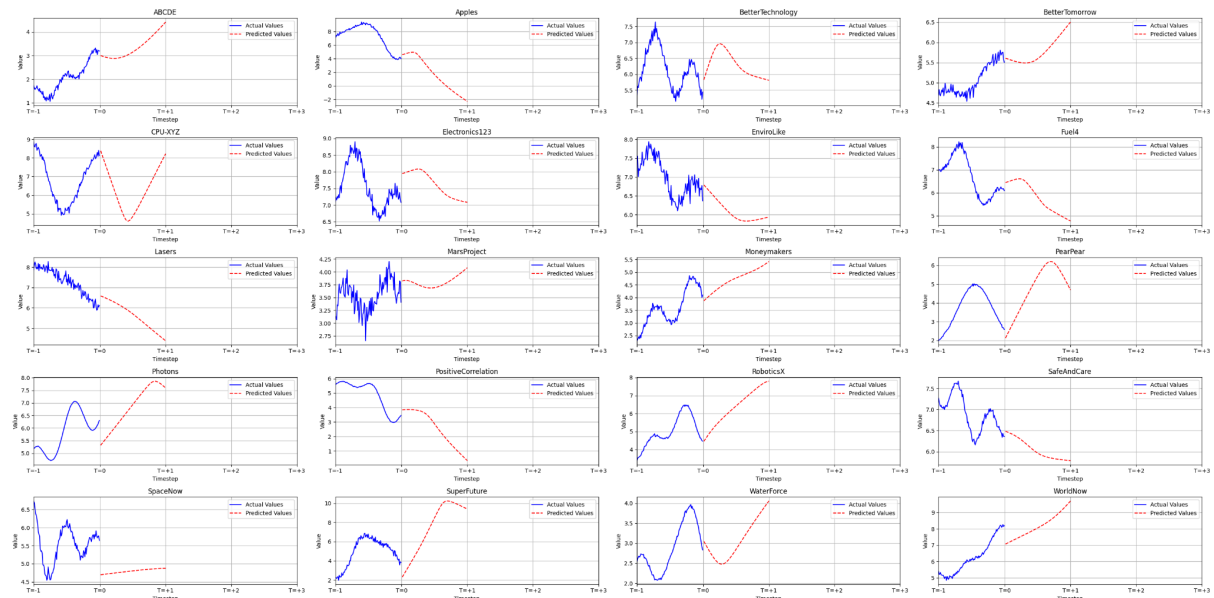


2.5 XGBoost



2.6 NODE

Neural Ordinary Differential Equation (NODE) is a state-of-the-art deep learning model that treats transformations of data as continuous processes, rather than discrete layers like in traditional neural networks. It can be effectively used in the context of time-series prediction as can be seen in the below figure.

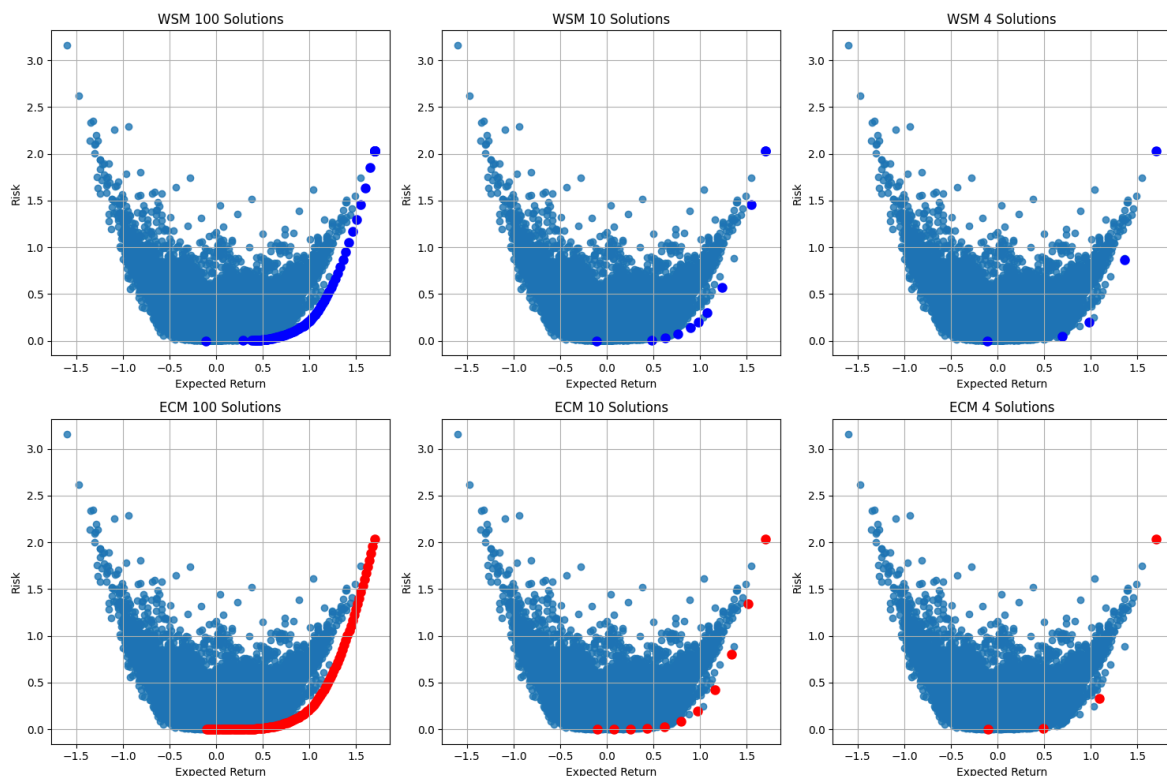


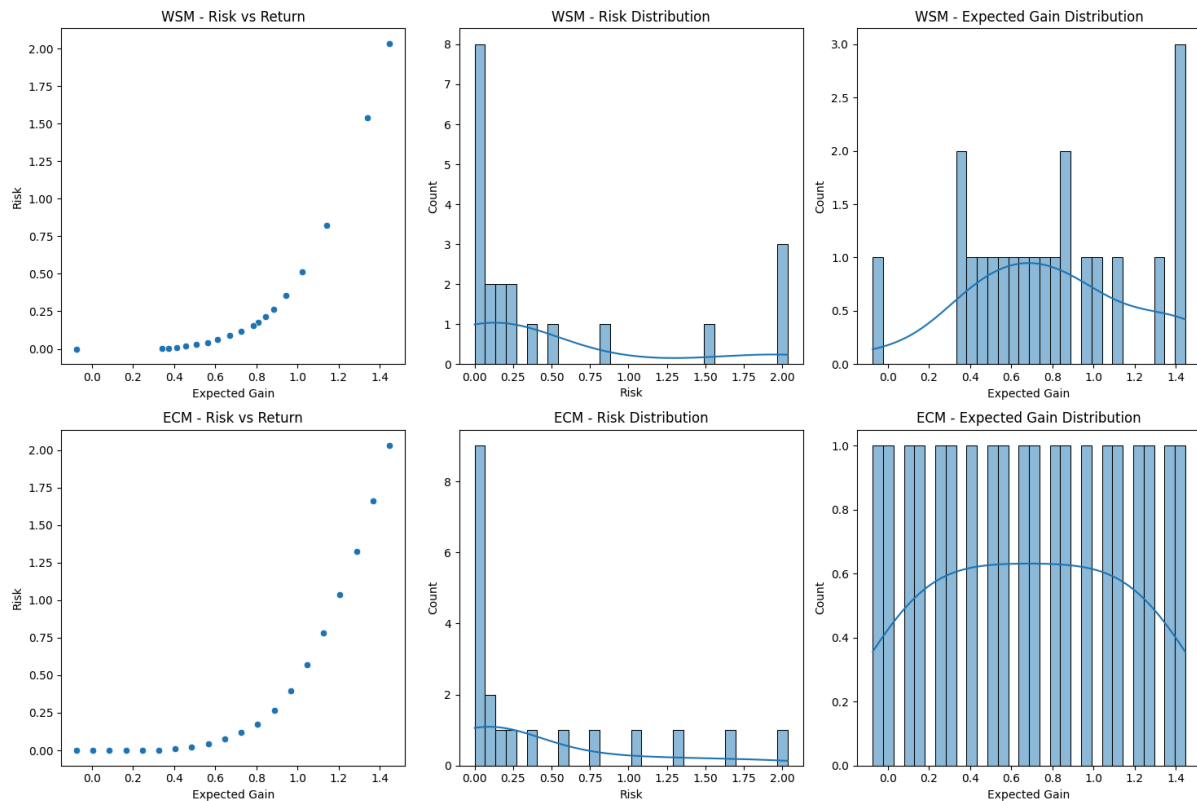
Conclusions

Many methods were either too simplistic to capture the true nature of the data (linear regression, ARIMA) or struggled with the small number of training examples (XGBoost, NODE). As such, no clear winner has appeared from the tests. Unfortunately, some of the most promising solutions like the sinusoidal regression functions ended up greatly surpassing the ranges of values produced by the original functions, potentially causing the model to be unstable. As such, we have decided to pick **NODE**'s predictions on which to base the solution, as it seems to capture some trends well, not greatly overestimating or underestimating the potential future values.

3. Pareto Front

To accomplish the task of Pareto Front estimation, two solutions were implemented that allowed us to find the Pareto front: the WSM and ECM methods. The WSM method is based on transforming a multi-objective optimization problem into a problem with a single objective through the use of weights, with varying weights allowing for the discovery of different points on the Pareto front. On the other hand, ECM uses thresholding to approximate the Pareto front. The below visualizations show the comparison of the discovered fronts for different approaches and for different number weight values/thresholds depending on the method. Additionally, a visualization comparing the risks, returns and repeated solutions can be seen. Based on both of the presented plots, it can be said that the ECM method seems to sample the front more uniformly, creating points that are more evenly spread, and not producing repetitions, unlike the WSM method which can produce the same solutions and samples the front around the convex shape of the figure produced by the uniform sampling of the available solutions much more frequently than for the other parts of the solution space.





4. Solution

In conclusion, ECM samples the front more uniformly, possibly providing a better approximation of the Pareto front and allowing the decision-maker to make a more informed judgment. However, the exact choice depends on the context of the problem, the exhibited preferences of the decision maker, and the trust placed in the regression model. In the end, when testing the data on the first stage of the “Portfolio Game” a solution with a relatively low risk but possibly over 111% total return was chosen thanks to its balance of expected returns and risk, however, the preference towards the risk was caused by the limited trust placed by us into the regression model due to few training examples available. As such, we have identified the solution: **[111.7229029981 0.0005054091 0.0000000244 0.0000000086 0.0000001247 0.0542921372 0.1761429231 0.0234277488 0.1583375966 0.0343588227 0.0059541302 0.0405949427 0.0163776905 0.0000000706 0.1603273564 0.0108636238 0.1572035995 0.0000001091 0.0048063507 0.0000010368 0.0914914603 0.0658202431]** as the best candidate for this problem. For a more thorough analysis, please refer to the project’s repository: <https://github.com/MatTheTab/MOO/tree/main>